



How do rising US interest rates affect developing East Asian economies?

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This Brief examines the impact of US interest rate hikes on developing East Asian economies. The effect will depend on the underlying factors driving the changes in US interest rates. US interest rates can be decomposed as stemming from three drivers: (1) developments in activity in the real economy; (2) inflation developments; and (3) reaction shocks, prompted by investors' assessments that the Federal Reserve has shifted toward a more hawkish or dovish stance. While rises in US interest rates driven by real activity developments have relatively benign effects on developing East Asian economies, those driven by inflation developments and reaction shocks adversely impact developing East Asian economies. The Brief can help policy makers in East Asian economies better understand the risks associated with the rises in US interest rates.

Introduction

Increases in US interest rates in response to rising inflation can negatively affect financial markets across the world and, subsequently, economic activity in developing economies, including in East Asia.

This Brief emphasizes the importance of comprehending the dynamic characteristics of the factors that influence shifts in US interest rates. With a more comprehensive understanding of the underlying drivers behind these changes, policy makers can make more informed decisions to face external disturbances.

The Brief follows Arteta, Kamin and Ruch (2022) and categorizes US interest rate changes as stemming from three drivers: (1) real activity developments (real shocks), which reflect changes in US interest rates that are prompted by changing prospects for US economic activity; (2) inflation developments (inflation shocks), which reflect changes in US inflation expectations; and (3) the monetary policy reaction function (reaction shocks), which reflect investors' assessment that the Federal Reserve's reaction function has changed (become more hawkish or dovish).

The findings suggest that US interest rate hikes due to real activity developments have relatively benign effects on East Asian economies. Real shocks are associated with higher external demand and better equity and sovereign risk outcomes, despite higher borrowing costs. However, US interest rate hikes due to inflation developments and shocks to the monetary policy reaction function adversely affect developing East Asian economies. These shocks lead to tighter financial conditions, which can reduce economic activity.

The results highlight the vulnerability of East Asian economies to the recent US interest rates hiking cycles that has been driven by a more hawkish Fed and higher inflation expectations. These findings underscores the need for policy makers in the region to monitor risks carefully and take appropriate measures to mitigate any negative effects on their economies.

Drivers of US interest rates

To decompose the shocks underlying changes in US interest rates, this study utilizes a sign-restricted Bayesian vector autoregressive (VAR) model with stochastic volatility, as in Arteta, Kamin, and Ruch (2022). The model utilizes monthly data spanning from January 1982 to December 2022 and includes four key variables: 2-year bond yields; 10-year bond yields; the S&P 500 index; and 5-year-ahead inflation expectations. The 10-year yield is included in addition to the 2-year yield to account for the persistent portion of inflation expectations and future monetary policy decisions.

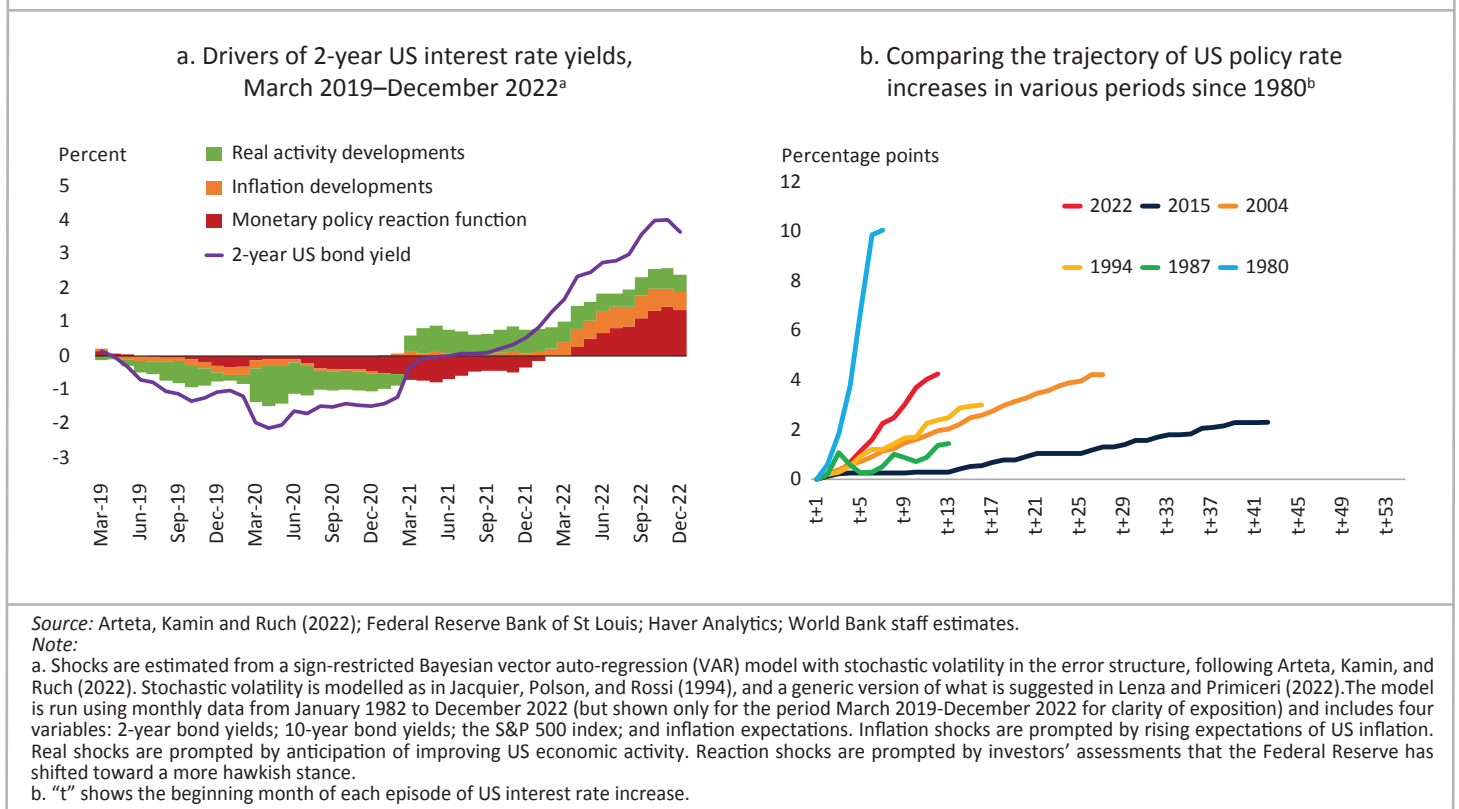
Increases in US interest rates can be attributed to several key drivers, including real activity developments (real shocks), inflation developments (inflation shocks), and the monetary policy reaction function (reaction shocks). Real shocks refer to changes in interest rates caused by fluctuations in US economic activity. When an interest rate increase is due to real shocks, it is accompanied by an increase in US yields, equity prices, and inflation expectations. Inflation shocks reflect changes in interest rates driven by shifts in inflation prospects. When an interest rate increase is due to inflation shocks, it results in an increase in US yields and an increase in inflation expectations, but a decrease in equity prices. Finally, reaction shocks are changes in interest rates driven by market perceptions of the Federal Reserve's reaction function. When an interest rate increase is due to reaction shocks, it results in an increase in US yields and a decrease in equity prices, similar to inflation shocks. However, unlike inflation shocks, reaction shocks are assumed to lower inflation expectations.

Real activity contributed positively to the US 2-year yield during 2021 (figure 1, panel a). Indeed, real shocks were the primary factor pushing the yield higher for most of 2021. As COVID-19 vaccination efforts gained momentum throughout the year, businesses reopened, and consumers resumed spending, economic activity picked up. These factors, coupled with ongoing fiscal support and the Fed's commitment to providing supportive monetary policy (including the announcement in mid-2020 of a change in its policy framework

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Figure 1. US interest rate developments



to target average inflation over a specific period), contributed to a surge in consumer demand. Shifts in monetary policy reaction helped push down the 2-year US yield during 2021, reflecting a perceived dovish shift in the Fed, a hold on changes in policy interest rates, and the uncertainty about the economic outlook and inflation developments (World Bank 2021).

In 2022, however, the factors influencing changes in the US 2-year yield underwent a notable shift. Initially, real activity developments were the primary driver of interest rate increases, but as the year progressed, inflation and monetary policy reaction began to play a dominant role. The trend reflected the Fed undertaking the fastest and steepest increase in policy rates since the late-1980s (figure 1, panel b).

Spillovers to developing East Asian economies

Given the evolution of factors that explain US interest rates movements, the impact of US monetary tightening on financial conditions and economic outcomes in developing East Asian economies is likely to have also changed. To assess the impact of these US interest rate shocks on financial and economic conditions in developing East Asia, local projection models following Jordà (2005) are estimated for China, Indonesia, Malaysia, the Philippines, and Thailand. The approach identifies impulse response functions through consecutive regression models at different horizons. Results are estimated six months ahead on monthly data from as far back as May 1998 to December 2022. Models include two lags and a dummy for the global financial crisis (January 2008–December 2009) and the COVID-19 pandemic (January 2020–December 2022). The findings for equity prices, exports, foreign exchange reserves,

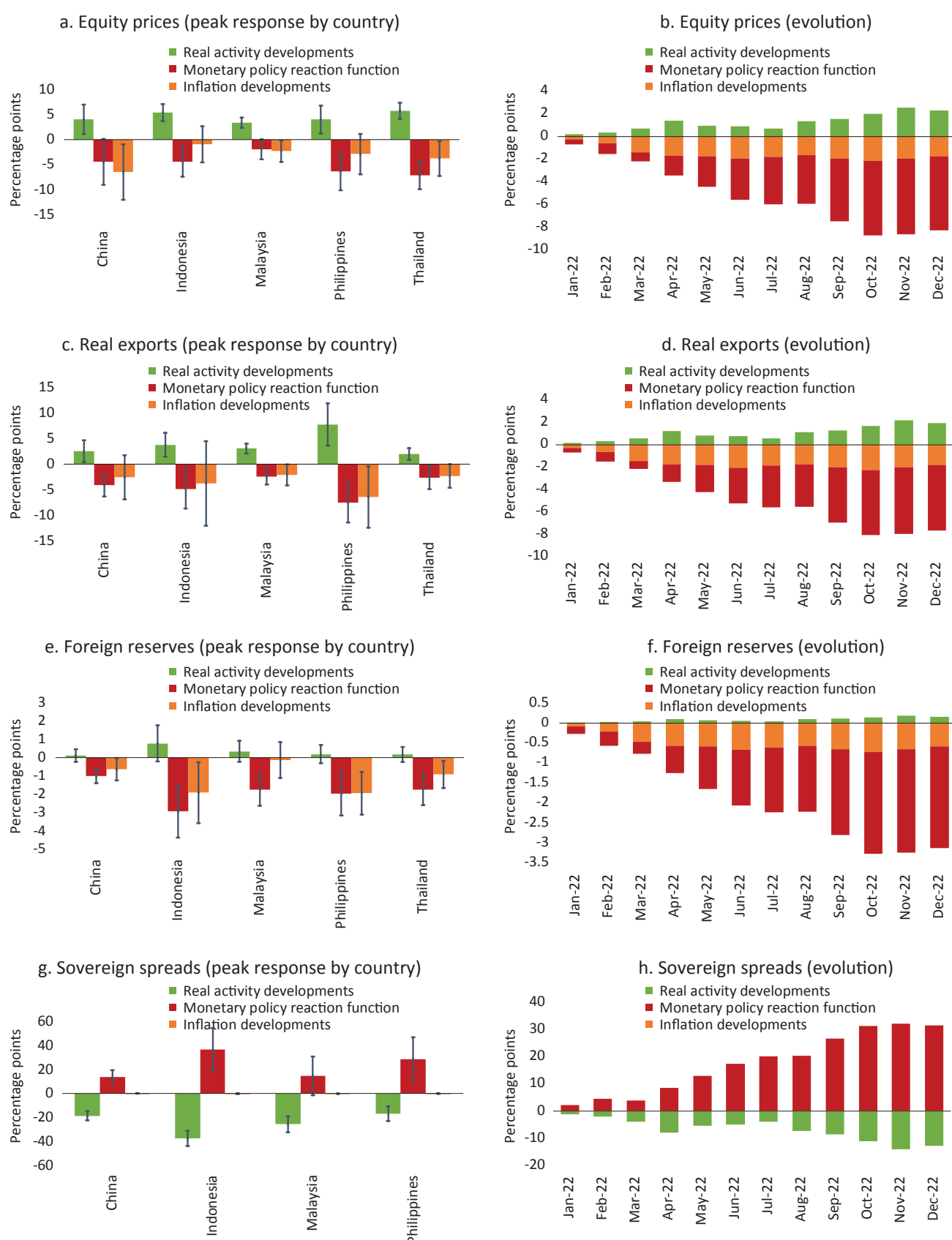
and sovereign risk spread are presented next and summarized in figure 2.

Increases in US interest rates driven by real activity developments (real shocks) had a positive effect on equity prices throughout 2022. Similarly, rises in US interest rates driven by real activity developments led to higher exports, reflecting the increase in demand, and lower sovereign risk spreads. They also appear to have had a negligible effect on reserves (figure 2).

Increases in US interest rates driven by inflation developments (inflation shocks), however, depressed equity prices in developing East Asian economies. Increases in US interest rates driven by inflation developments are associated with lower growth of goods exports due to lower external demand and led to loss of foreign reserves in developing East Asia economies.

Increases in US interest rates driven by monetary policy reaction function (reaction shocks) led to a decrease in equity prices, a decrease in real goods exports, loss of foreign currency reserves, and higher sovereign risk spreads. Overall, over the course of 2022, the net effect on developing East Asian economies of the different factors driving US interest increases was negative. For example, a 25-basis point increase in the 2-year US yields depressed equity prices by about 6 percent in developing East Asia at the end of 2022 compared to the previous year. While real activity developments in the United States helped boost equity prices in the region, these gains were overshadowed by declines driven from inflation developments and the monetary policy reaction function in the United States.

Figure 2. Impact of US interest rate yields on economic activity in developing East Asia



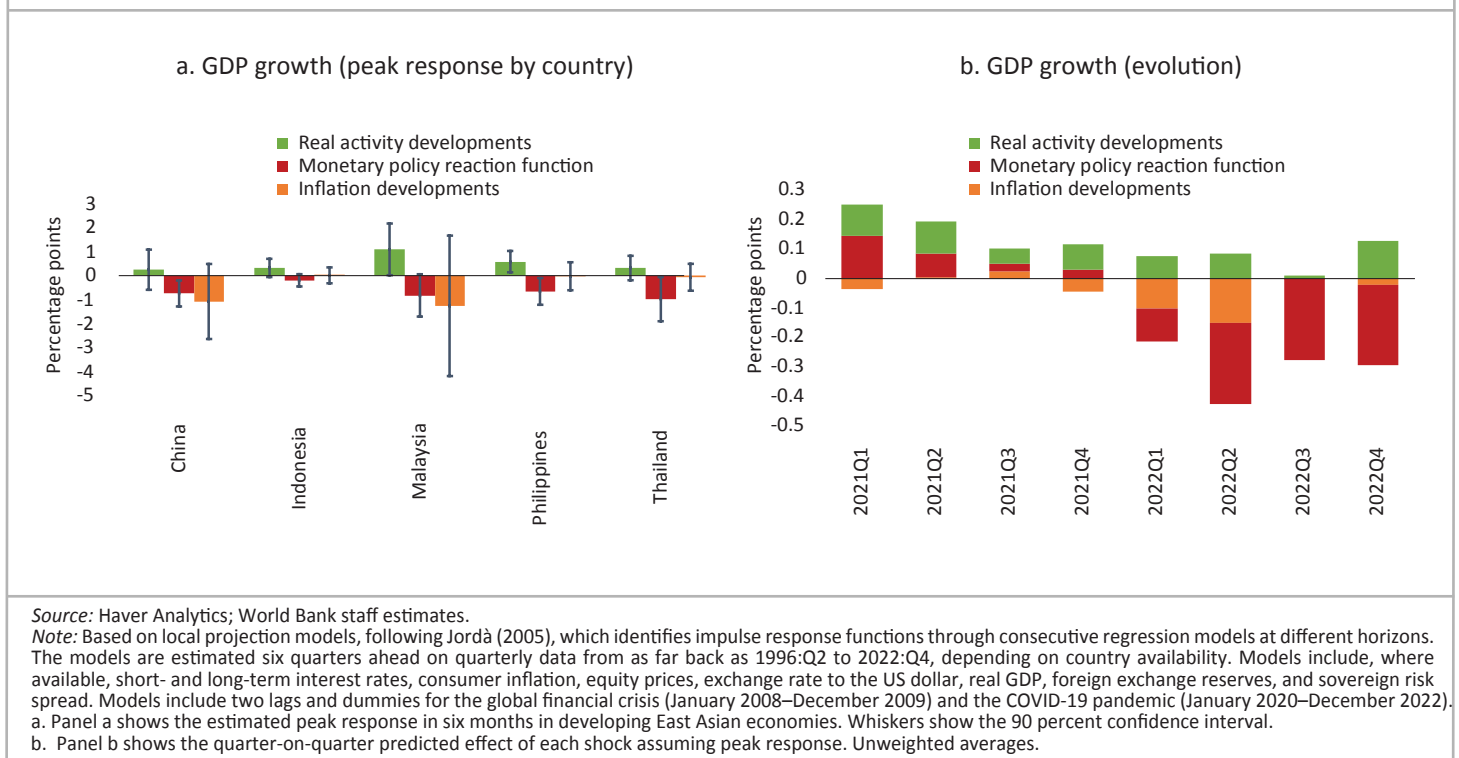
Source: Haver Analytics; World Bank staff estimates.

Note: Based on local projection models, following Jordà (2005), which identifies impulse response functions through consecutive regression models at different horizons. The models are estimated six months ahead on monthly data from as far back as May 1998 to December 2022, depending on country availability. Models include, where available, short- and long-term interest rates, consumer inflation, equity prices, exchange rate to the US dollar, real exports in US dollars, foreign exchange reserves, and sovereign risk spreads. Models include two lags and dummies for the global financial crisis (January 2008–December 2009) and the COVID-19 pandemic (January 2020–December 2022).

a. Panels a, c, e, and g show the estimated peak response in six months in developing East Asian economies. Whiskers show 90 percent confidence interval.

b. Panels b, d, f, and h show the predicted effect of each shock at peak response. Unweighted averages.

Figure 3. Impact of US interest rate yields to GDP in developing East Asian economies



Increases in US interest rates during 2022 also hurt GDP growth in developing East Asian economies. While real shocks are associated with higher GDP growth, especially in Malaysia and the Philippines, inflation shocks and especially monetary policy reaction shocks depressed GDP growth. Increases in interest rates due to monetary policy reaction shocks are associated with a significant decline in emerging market investment and private consumption (Arteta, Kamin, and Ruch 2022). In contrast, real shocks lead to higher real exports. The rise in US interest rates in 2022 depressed quarter-on quarter GDP growth by 0.2 percentage points, on average (figure 3).

Policy relevance

The findings of this study demonstrate that developing East Asian economies are susceptible to the impact of US interest rate hikes. While policy makers in the region should remain vigilant and closely monitor financial tightening in advanced economies, the response should be cautious and based on rigorous analysis. When increases in US interest rates are driven by improving demand, then nations in the region could see some positive effects on their economic activity.

But interest rate hikes in the United States can lead to a reduction in investment, consumption, and exports in East

Asian economies and significantly tighten financing conditions. These adverse effects are more likely when inflation developments and monetary policy reaction are driving the increase in interest rates.

It is essential for policy makers to take appropriate measures to mitigate any negative effects of interest rate hikes on the East Asian economies. They need to remain proactive in identifying and managing the risks associated with such shocks. Additionally, policy makers must develop and implement policies that are robust enough to withstand the impact of interest rate hikes. This can include measures such as increasing foreign exchange reserves, implementing flexible exchange rate regimes, and introducing policies to enhance financial stability.

In conclusion, the vulnerability of East Asian economies to US interest rate hikes driven by factors other than improving demand is a cause for concern. Policy makers in the region must remain vigilant and take proactive measures to mitigate any adverse effects on their economies. By doing so, they can help safeguard the long-term economic growth and stability of the region.

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